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Exploration of driver stress when resuming control from highly automated driving in an emergency situation

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ABSTRACT

Introduction: In a highly automated vehicle, the resumption of manual control often takes place in emergency situations automation cannot handle. Time pressure may increase driver stress and lead to risky behaviors. For safety purposes, this study examines the impacts of emergency takeovers on driver stress.
Method: The impacts of emergency takeovers on driver stress were explored through heart rate, pupil diameter and perceived stress, gathered from 36 drivers in a driving simulator. Three degrees of emergency were used: non–, moderate, and extreme emergency.
Results: The results indicate a pupil dilation, high levels of perceived stress and a long-lasting increase in heart rate when drivers had to resume control in moderate and extreme emergency situations. The results also show a lower perceived stress level and a bi-phasic cardiac pattern—composed of a short-lasting decrease in heart rate prior to long-lasting increase in heart rate—when drivers were warned to regain control (moderate emergency) compared to when they were not warned (extreme emergency).
Conclusion: The results (1) suggest that time urgency is a key factor in the induction of driver stress during takeovers, (2) support the existence of a warning-induced preparatory response—characterized by a short-lasting decrease in heart rate—and previously observed in research on defensive behaviors, and (3) support the hypothesis that the preparatory response is a means of reducing driver stress. Overall, the results provide valuable information about cognitive preparation for emergency takeovers, and more generally about driver stress in emergency situations.

1. Introduction

1.1. Emergency takeovers and driver stress

The automotive industry is currently undergoing a major transition. Indeed, the driver is gradually pushed out of the control loop and replaced by driving automation. However, for all situations that automation cannot manage, the driver is currently asked to regain manual control of the vehicle. Ironically, the resumption of manual control often takes place in emergency situations. Under

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emergency situations, drivers have a restricted amount of time to prepare to resume control, which means little time to process information and handle the vehicle trajectory correctly (Stahl et al., 2014). Time constraints in emergency situations may result in the perception that internal resources are insufficient to accomplish the task effectively. Given that driver stress has been related to poor driving performances and risky behaviors (Ge et al., 2014; Matthews et al., 1998; Hancock and Desmond, 2001; Rendon-Velez et al., 2016), the resumption of manual control in emergency situations is a major safety concern (Navarro, 2019). For safety purposes, it is essential to better understand driver stress during emergency takeovers.

1.2. Defensive behaviors: A theoretical approach to understand driver stress during emergency takeovers

The vast literature on defensive behaviors has reported a relationship between active preparation (or anticipation) under threat imminence and cardiac deceleration (Gladwin et al., 2016; Hashemi et al., 2019; Löw et al., 2015; Rösler and Gamer, 2019). The cardiac acceleration following the cardiac deceleration can be viewed as a stress response (Cannon, 1915) that allows the individual to flee or fight the threat (Fanselow, 1994; Lang et al., 1997). Therefore, examining drivers' cardiac responses in the light of defensive behavior would provide valuable information about cognitive preparation for imminent takeovers, and more generally about driver stress in emergency situations.

1.3. Using a general framework to study the impacts of emergency takeovers

In the highly automated driving context, Lu et al. (2016) proposed a framework that classifies the transitions in which the human-automation system changes from one driving state to another. This framework defines the transitions in terms of what the driver and the automation system are doing, not what they should be doing. Among the transitions defined, three of them are takeovers, i.e. they engage a shift of control from the automation system. These three transitions, distinguished by the optional or mandatory nature of the takeover, as well as by the agent that initiated the transition (driver or automation), are: Optional Driver-Initiated transition (ODI), Mandatory Automation-Initiated transition (MAI) and Mandatory Driver-Initiated transition (MDI). A parallel can be drawn between the three takeovers defined in the framework of Lu and colleagues (2016) and three degrees of emergency situations in which they are involved. Indeed, an ODI transition would be initiated by a driver most of the time in a non-emergency situation since both agents (driver and automation) are able to drive. Conversely, MAI and MDI transitions would be initiated in emergency situations since for both, the automation system is no longer able to drive. In the MAI transition, the automation system detects its own inability to drive and thus demands that the driver regain control of the vehicle via a takeover instruction. In contrast, the MDI transition is triggered by the driver when he/she detects the automation system's inability to drive the vehicle. In other words, in the MDI transition, no takeover request was provided by the automation system and the driver regained control based solely on his/her own assessment of the system's incapacity. Because of the lack of warning (i.e. takeover request), the MDI transition should reasonably be associated with a higher degree of emergency than the MAI transition. Apart from the agent that initiated the transition (driver or automation) and the optional or mandatory nature of the takeover, the overall complexity of the driving situation during takeover should also significantly influence the urgency of the situation. For example, a driving situation involving potential collisions with pedestrians, thus affecting peer safety, should generate greater stress levels and more urgency than a situation with potential collisions that does not involve pedestrians.

As a result, the current study aimed to explore the impacts of emergency takeovers on driver stress by manipulating emergency using (1) the agent that initiated the transition (driver or automation), (2) the optional or mandatory nature of the takeover, and (3) the complexity of the driving situation (e.g., involving risks of collision with or without pedestrians). From these manipulations, three degrees of emergency takeovers were derived: non–, moderate, and extreme emergency.

We had the following specific hypotheses:

Hypothesis 1.

- Based on a general hypothesis that the degree of emergency in driving situations determines stress levels, we expected to observe more stress when drivers regained control in an emergency situation than in a non-emergency situation.

Hypothesis 2.

- Based on the theory of defensive behaviors, the presence of a warning (i.e., take-over request) in an emergency situation would offer the driver additional time for active preparation (or anticipation). We assumed that active preparation would mitigate the urgency of the situation, and thus stress levels. As a result, we expected less stress when drivers had to regain control in a warned emergency situation (i.e. moderate emergency) than in a non-warned emergency situation (i.e. extreme emergency).

In this study, the pupil diameter was used as an indicator of drivers' cognitive processing of the situation, while heart rate was used as a sensitive measure of driver stress (Keraustret et al., 2021). In addition, self-assessments were used to subjectively estimate the stress responses.

2. Materials and methods

2.1. Participants

Thirty-six participants (thirteen women, twenty-three men) aged between 20 and 59 years (M = 33.21; SD = 11.63) with at least one year of driving experience (M = 13.10; SD = 10.22) were recruited. All participants had normal or corrected-to-normal vision and declared no cognitive disorders and no heart disease. All participants signed an informed consent form stating that electrocardiogram signals, ocular and driving information would be collected throughout the driving simulator experiment.

2.2. Study design and procedure

Participants started the experiment with an 8-minute training driving session to get familiar with the driving simulator and the use of the automation system. During the training, participants were free to take back control and delegate to the automation system at any time. Afterwards, participants completed two sessions in a counterbalanced order: the *takeover session* and the *monitoring session*. In these sessions, drivers were instructed, first, to monitor the vehicle's operations when the automation system was in control, and second, to regain control when the system requested it explicitly or when drivers suspected an automation failure.

During the takeover session, participants were prompted to regain control from automation in three different driving situations (see Fig. 1):

1. The first driving situation involved a low-traffic toll area approached at low speed, making it a relatively simple situation with a low risk of collision. The situation was also marked by the display on the dashboard of a warning from the automation system just before entering the toll zone. The warning indicated to the drivers that they could, if they wanted to, regain control of the vehicle ("*Do you want to take over control?*") and there was no time limit to do so. Therefore, in the case of late takeovers, there were no negative consequences since the automation system handled the driving correctly. After resuming control of the vehicle, the driver simply had to drive through the automatic toll gate without stopping. This transition, also referred to as an Optional Driver-Initiated transition (ODI), should not be confused with an automation-initiated transition since the driver was free to resume control of the vehicle or not (Lu et al., 2016). Nonetheless, to encourage the recovery of vehicle control, and thus enable the analysis of the ODI transition, the drivers were also instructed, before the driving sessions, that the suggestion by the automation system reflected a low confidence in its ability to control the vehicle flawlessly in specific situations. Despite this incentive, five participants (in

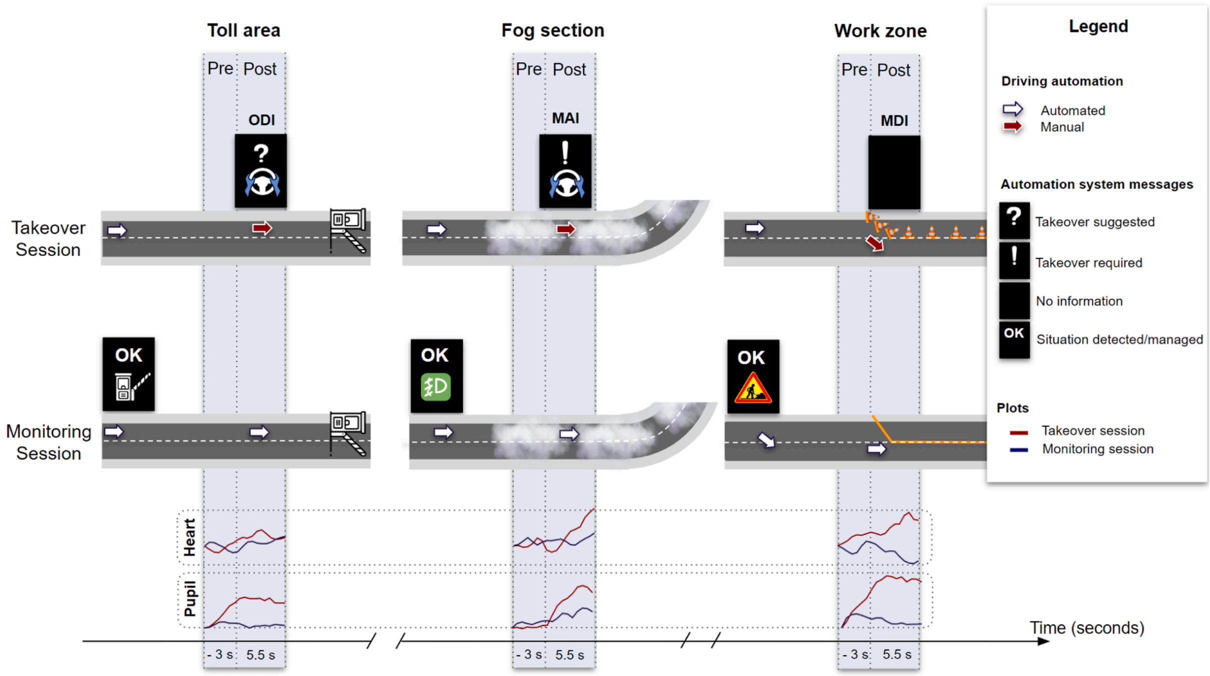

Fig. 1. Overview of the three driving situations (toll area, fog section, and work zone) in the takeover and monitoring sessions. The takeover session included the Optional Driver-Initiated transition (ODI), Mandatory Automation-Initiated transition (MAI), and Mandatory Driver-Initiated transition (MDI). For each driving situation, the mean changes in heart rate and pupil diameter in the takeover session were compared to those in the monitoring session, and analyzed over two time windows (pre-takeover and post-takeover). The x-axis represents the measurement period for both time windows.

addition to the number reported in the participants section) had to be excluded from analyses because they did not regain manual control in that situation.

2. The second driving situation was relatively complex. In this situation, an intense fog, a heavy traffic and a complex turning maneuver were simultaneously involved. Before the turning maneuver, the driver was also involved in a Mandatory Automation-Initiated transition (MAI). To trigger this transition, a warning, displayed on the dashboard, was issued by the automation system to instruct the driver to urgently regain control of the vehicle ("*Take over control!*"). Under this message, the driver was also informed of the reason for this request, namely the blinding of the vehicle's sensors by the fog. As a result, the driver had to regain control of the vehicle as quickly as possible and guide it through a complex maneuver with poor visibility while taking into account the surrounding vehicles. The driving scenario was created so that late takeovers resulted in collisions with surrounding vehicles and road infrastructure.
3. The last driving situation was highly complex since it involved a work zone, heavy traffic and road workers to be avoided. Specifically, in this situation, the automation system was programmed so that the vehicle keeps following the lane markings, despite a conflict of information with the traffic cones intended to guide the traffic to a temporary lane. The driving scenario was designed in such a way that the automation system did not inform the driver of the vehicle's decision regarding its direction, i.e., either to follow the road markings or the traffic cones. Therefore, the driver was involved in a Mandatory Driver-Initiated (MDI) transition since he or she had to identify, without the aid of any explicit warning, the failure of the automation system to guide the vehicle in the right direction (i.e. the temporary lane). In addition, late takeovers after the awareness of a failure resulted in collisions with first traffic cones, then stationary vehicles of highway maintenance and pedestrians.

For each of the three transitions (i.e, ODI, MAI and MDI), the takeover was initiated by pressing a pedal (throttle or brake) and the effects of the driving situation were studied over the 3 s preceding the takeover (pre-takeover period), while the impact of the transition was investigated over the 5.5 s following the takeover (post-takeover period). The time windows for the pre- and post-takeover periods were retained so that the duration was long enough to study the effects of the transition and the driving situation and short enough to not be impacted by the road conditions that changed as the driver moved through the environment.

During the monitoring session, participants were faced with the same three driving situations as during the takeover session, however, they were never invited to resume control of the vehicle. This session was thus designed to provide control conditions for the three transitions presented in the takeover session. Therefore, in the monitoring session, the automation system informed the driver that the situation was fully under control as soon as the driving situations were initiated so as to avoid untimely takeovers.

For each driving situation, data from heart rate and pupil diameter were gathered and compared between the takeover session and

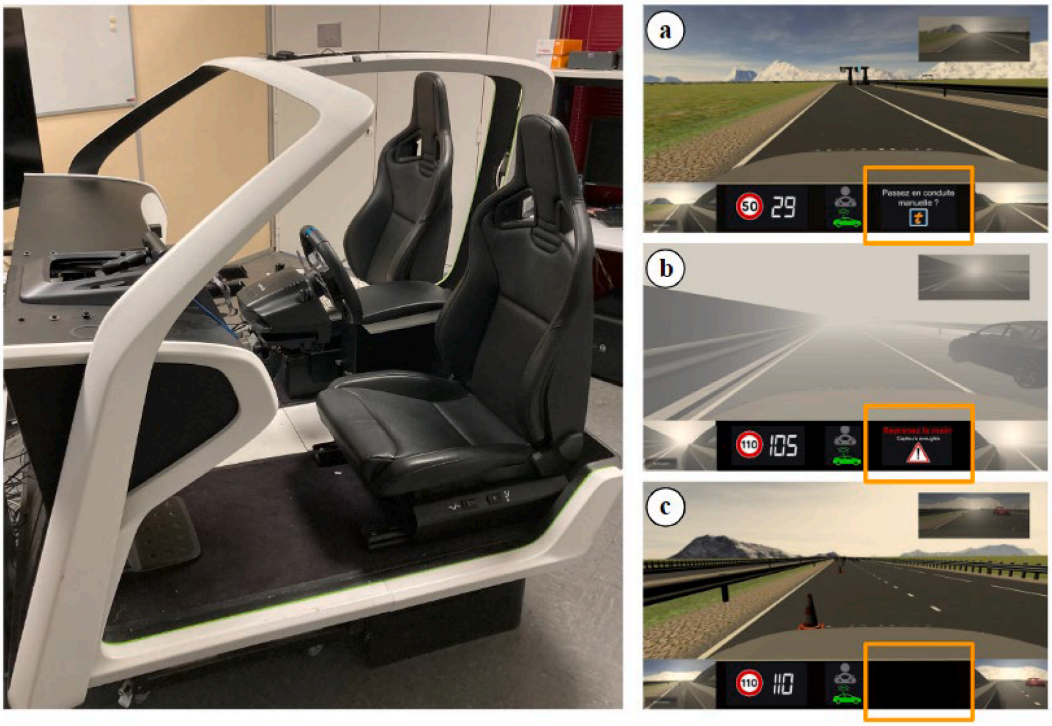

Fig. 2. Overview of the homemade driving simulator and driving scenarios designed on Unity 3D software: (**a**) a takeover suggestion ("*Do you want to take over control?*") from the automation system was delivered to trigger an ODI transition, (**b**) a takeover request ("*Take over control!*") from the automation system was delivered to trigger a MAI transition, (**c**) no takeover request from the automation system was delivered to trigger a MDI transition.

the monitoring session. Before and at the end of the experiment, drivers were asked to report perceived stress at rest and for the three driving situations experienced both in the monitoring and the takeover sessions. Perceived stress was assessed using a 5-point Likert stress scale, ranging from 1 “not stressful at all” to 5 “extremely stressful”.

2.3. Apparatus

The experiment was performed in a homemade, fixed-base driving simulator (see Fig. 2) with a fully equipped interior: automatic gearbox, steering wheel and pedals (Logitech G29). Simulation environment, rear-view mirror and side mirrors were displayed on a screen (horizontal field of view: 130°) in front of the simulator. Unity 3D software was used to design the simulated driving environment. A physiological data acquisition system (BIOPAC MP160) was set up to collect drivers' cardiac responses with a sampling rate of 500 Hz. In order to collect cardiac data, two electrodes were placed on the manubrium of the sternum and the left lower rib cage, while the reference electrode was placed on the driver's right side at the top of the hip. Drivers' ocular responses were collected via an eye tracker (FOVIO) with a sampling rate of 60 Hz. A calibration was performed for each driver prior to the training drive. Finally, Rtmaps software was used to timestamp, record, and synchronize the data from the different sensors.

2.4. Data processing

Perceived stress. Perceived stress scores were normalized for each driver by computing the z-score as follows: $z = (x - \mu) / \sigma$, where:
z = the standard score for a random driver,
x = the reported score by the driver,
μ = the mean of all reported scores by the driver (at rest, in takeover session, in monitoring session), and.
σ = the standard deviation of all reported scores by the driver (at rest, in takeover session, in monitoring session).

Heart. Using the same methodology as previous research (Kerautret et al., 2022, Pepin et al., 2017), cardiac data were pre-processed as follows: (1) a band-pass filter between 2 and 40 Hz was applied on the electrocardiogram signals to remove signal drifts, (2) R-R peaks were detected using an automatic detection procedure (AcqKnowledge 5.0 software) then extracted after a visual check of the electrocardiogram signals for artifacts and correction for misplaced or omitted R waves, (3) the R-R intervals were converted into heart rate (beats per minute), (4) the heart rate data was sampled every 0.5 s for periods of −3 up to 0 s and 0 up to 5.5 s after takeover. The heart rate value corresponding to the moment the driver pressed the pedal was used as a baseline for the post-takeover heart rate values. Three seconds prior to the moment the driver touched the pedal was used as a baseline for the pre-takeover heart rate values. Data sampling was performed using a cubic spline interpolation. (5) Finally, heart rate change was calculated by subtracting the baseline from each post-baseline heart rate value.

Pupil. Pupil data pre-processing consisted of: (1) averaging the data from both eyes to obtain one value for each participant at any moment, (2) sampling pupil data every 0.5 s over two time windows (pre-takeover and post-takeover) in the same way as for heart rate, (3) normalization of the pupil data for each driver, according to her or his average pupil diameter over the three drives, (4) subtracting the baseline value (i.e. when driver touched the pedal) from each post-baseline pupil data value.

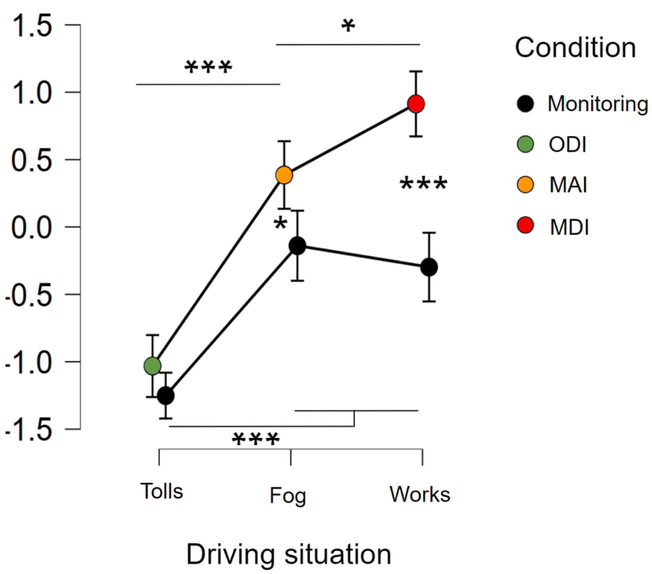

Fig. 3. Mean perceived stress levels for the takeover conditions (i.e., ODI, MAI, MDI) and the monitoring conditions in each driving situation (toll, fog, works). The vertical bars denote the 95 % confidence intervals. * $p < .05$. *** $p < .001$.

2.5. Statistical analysis

Statistical analysis was performed using JASP software (version 0.14.0.0). For all the analyses, the significance level α was set at $p < .05$.

Perceived stress assessments. Perceived stress was assessed by performing a 2 × 3 repeated-measures ANOVA including the condition (takeover vs monitoring) and driving situation (toll, fog, works) as within-subject factors. Effect sizes were estimated using partial eta-squared (η_p^2). Post-hoc comparisons were run by applying Bonferroni corrections.

Physiological responses. To evaluate driving situation-related responses, we carried out a 2 × 6 repeated measures ANOVA, for each of the three situations, over the pre-takeover period. The condition (monitoring vs takeover) and time point (6 levels) were included as within-subject factors. Then, to evaluate takeover-related responses, we performed a 2 × 11 repeated measures ANOVA, for each situation, over the post-takeover period. Likewise, the condition (monitoring vs takeover) and time point (11 levels) were included as within-subject factors. For all analyses, degrees of freedom were adjusted according to the Greenhouse-Geisser method to compensate for violations of the sphericity assumption. Effect sizes were estimated using partial eta-squared (η_p^2). ANOVAs indicating a significant condition × time interaction led to pairwise comparisons for each time point between the monitoring and the takeover conditions.

3. Results

3.1. Perceived stress assessments

In order to assess whether perceived stress differed across conditions and driving situations, we performed a repeated-measures ANOVA evaluating the influence of condition (monitoring vs takeover) and driving situation (fog vs toll vs works) (Fig. 3). A main

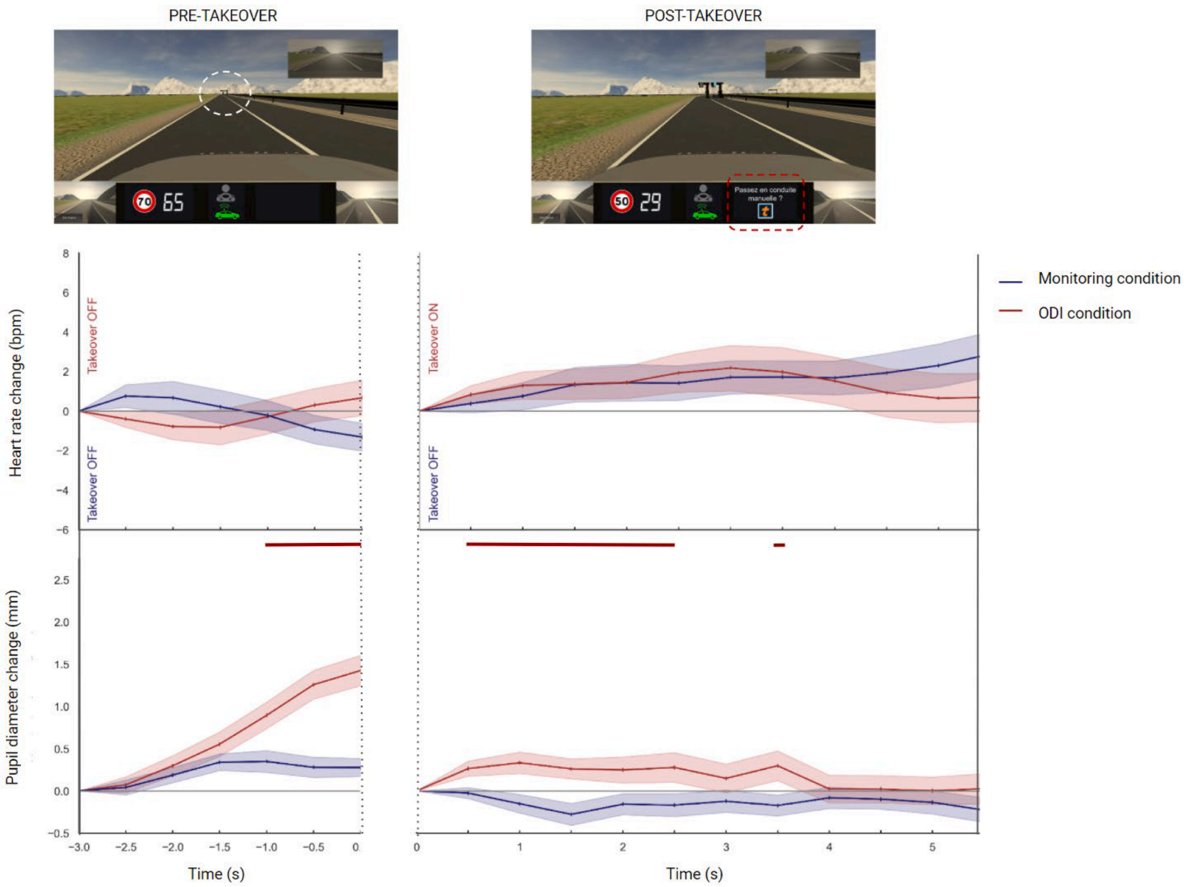

Fig. 4. The pictures depict the simulated environment 3 s before the takeover (left side), arriving at a toll gate (white dotted area), and at the moment when the takeover is suggested (right side) with a warning (red dotted box). The plots display changes in heart rate and pupil diameter prior to the takeover (pre-takeover period) and from the takeover (post-takeover period). The signals in blue were recorded during the monitoring condition, and in red during the ODI takeover condition. Shaded areas denote standard errors of the mean. Time points with significant differences between the monitoring condition and the takeover condition are displayed at the top of each figure as a red horizontal line. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

effect of condition ($F_{(1, 35)} = 37.93$, $p < .001$, $\eta_p^2 = 0.517$), a main effect of driving situation ($F_{(2, 10)} = 99.97$, $p < .001$, $\eta_p^2 = 0.741$), and a significant interaction between condition and driving situation ($F_{(2, 70)} = 9.95$, $p < .001$, $\eta_p^2 = 0.221$) were found.

Post-hoc analyses indicated more perceived stress in the MDI and MAI conditions than in the corresponding monitoring conditions (MDI, $t = 7.16$, $p < .001$, MAI, $t = 3.103$, $p < .05$). In contrast, the perceived stress levels were not significantly different between the ODI condition and the corresponding monitoring condition (NS). In addition, the MDI condition revealed a higher perceived stress level than in the MAI condition ($t = 3.31$, $p < .05$), which was in turn higher than in the ODI condition ($t = 8.89$, $p < .001$). Within the monitoring condition, the fog area and the work zone situations highlighted both higher perceived stress levels than in the toll area situation (fog, $t = 6.97$, $p < .001$, works, $t = 5.98$, $p < .001$). However, no significant difference was observed between the monitoring situation in the fog area and the work zone (NS).

3.2. Physiological responses

In order to explore driving situation-related cardiac responses, we ran a 2 × 6 repeated measures ANOVA for each of the three driving situations, including the within-subject factors condition (monitoring vs takeover) and time point (6 levels), over the pre-takeover period. Then, to investigate takeover-related cardiac responses, we performed a 2 × 11 repeated measures ANOVA, for each of the three situations, including the within-subject factors condition (monitoring vs takeover) and time point (11 levels), over the post-takeover period. Finally, we repeated the same process with pupil diameter.

3.2.1. ODI condition

Heart. The ANOVAs revealed no significant differences for heart rate between the ODI condition and the monitoring condition over both the pre- and post-takeover periods, (see Fig. 4, and Table 1 for ANOVA statistics).

Pupil. Differences in pupil diameter were observed between the ODI condition and the monitoring condition over the pre- and the post-takeover periods. Indeed, significant condition × time interactions were notably revealed in the pre-takeover period [$F_{(3.14, 97.41)} = 13.1$, $p < .001$, $\eta_p^2 = 0.297$] and in the post-takeover period [$F_{(5.28, 163.7)} = 2.7$, $p < .05$, $\eta_p^2 = 0.080$] (see Table 1). Then, pairwise comparisons indicated that pupil diameter was significantly greater in the ODI condition than in the monitoring condition from 1 s in the pre-takeover period, and from 0.5 s to 2.5 s and at 3.5 s in the post-takeover period (see colored lines above the graph in Fig. 4**,** and Appendix for pairwise comparisons in details).

3.2.2. MAI condition

Heart. The ANOVA revealed no significant differences for heart rate over the pre-takeover period between the MAI condition and the monitoring condition. However, in the post-takeover period a significant condition × time interaction was noticed [$F_{(3.69, 129.2)} = 8.2$, $p < .001$, $\eta_p^2 = 0.190$] (see Fig. 5, and Table 2 for ANOVA statistics). Pairwise comparisons then informed that the MAI condition differed significantly from the monitoring condition, being lower at 1.5 s and greater at 5.5 s after that drivers initiated the takeover (see colored lines above the graph in Fig. 5**,** and Appendix for pairwise comparisons in details).

Pupil. Regarding pupil diameter, no differences were observed between the MAI condition and the monitoring condition in the pre-takeover period (see Table 2). Conversely, in the post-takeover period, a condition × time interaction was clearly identified [$F_{(3.88, 136.0)} = 9.9$, $p < .001$, $\eta_p^2 = 0.221$], indicating that pupil diameter was wider from 1 s to 5.5 s in the MAI condition than in the monitoring condition (see Fig. 5**,** and Appendix).

3.2.3. MDI condition

Heart. The ANOVA found no differences in heart rate change between the MDI condition and the monitoring condition in the pre-takeover period (see Fig. 6, and Table 3 for ANOVA statistics). In contrast, in the post-takeover period, a significant condition × time interaction was reported [$F_{(2.40, 84.16)} = 10.5$, $p < .001$, $\eta_p^2 = 0.231$]. Pairwise comparisons then specified that heart rate was greater in the MDI condition compared to the monitoring condition from 0.5 to 5.5 s after the takeover (see colored lines above the graph in Fig. 6, and Appendix for pairwise comparisons).

Pupil. The ANOVAs revealed significant differences for pupil diameter between the MDI condition and the monitoring condition over both the pre- and post-takeover periods. Indeed, significant condition × time interaction was found [Pre-takeover, $F_{(2.94, 103.0)} = 25.7$, $p < .001$, $\eta_p^2 = 0.424$, Post-takeover, $F_{(4.24, 148.4)} = 5.0$, $p < .001$, $\eta_p^2 = 0.127$] (see Table 3). Subsequent pairwise comparisons indicated that the pupil diameter was higher in the MDI condition than in the monitoring condition, from 1.5 s to 5.5 s over the pre-takeover period, and from 0.5 s to 5 s over the post-takeover period (see Fig. 6, and Appendix).

Table 1
ANOVA outcomes comparing the ODI condition with the monitoring condition, in the pre- and post-takeover periods.

Measure	Period	Condition			Time			Condition x Time		
		F	(df)	η^2p	*F*	(df)	η^2p	*F*	(df)	η^2p
Heart rate	Pre-takeover	0.16	(1, 35)	0.005	0.45	(2.61, 81.07)	0.014	2.7	(2.33, 72.31)	0.082
	Post-takeover	0.023	(1, 35)	0.000	2.1	(3.12, 96.93)	0.063	0.69	(2.11, 65.63)	0.022
Pupil diameter	Pre-takeover	**8.9****	(1, 31)	0.225	**25.8*****	(2.90, 89.89)	0.455	**13.1*****	(3.14, 97.41)	0.297
	Post-takeover	**5.3***	(1, 31)	0.148	1.0	(4.43, 137.5)	0.032	**2.7***	(5.28, 163.7)	0.080

Note. F (df) = statistics of ANOVA and degrees of freedom (in parentheses). * *p* <.05. ** *p* <.01. *** *p* <.001.

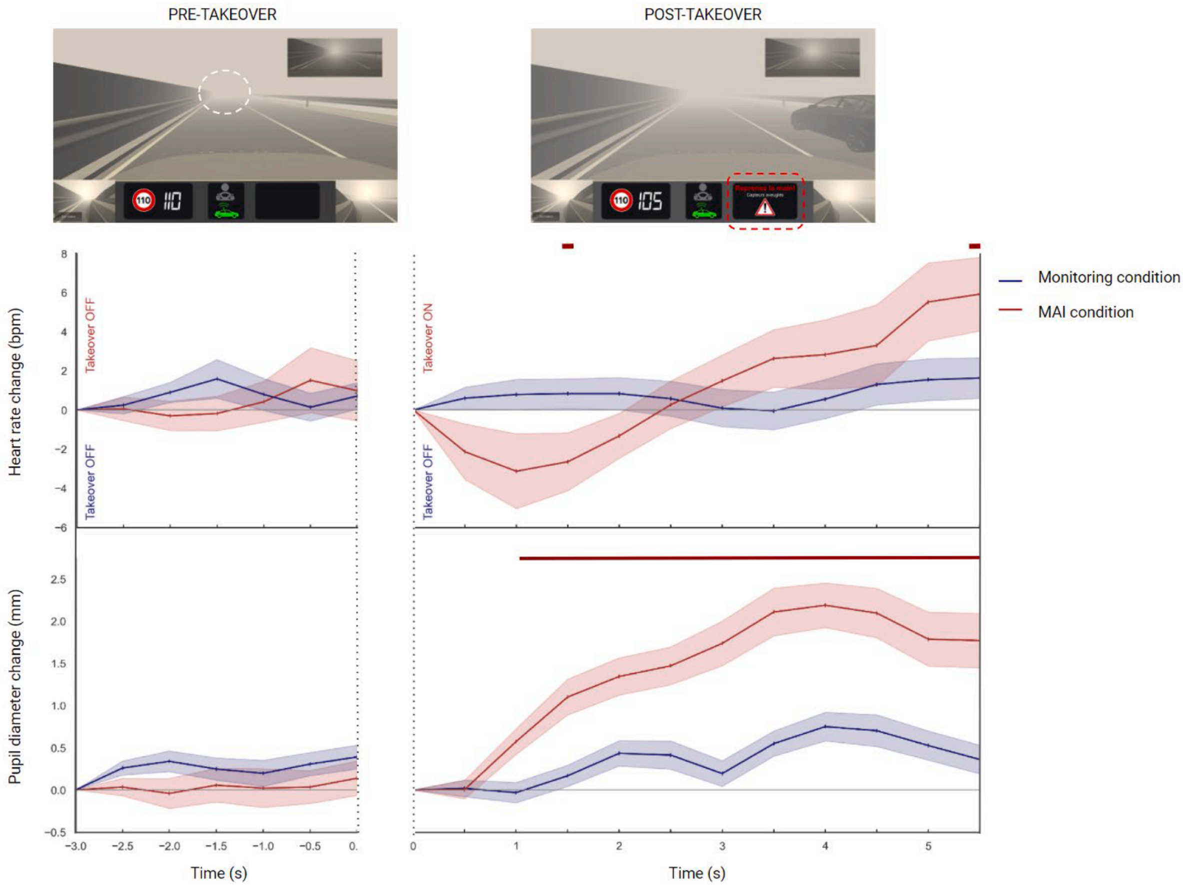

Fig. 5. The pictures depict the simulated environment 3 s before the takeover (left side), in the middle of a fog (white dotted area), and at the moment when the takeover is requested (right side) with a warning (red dotted box). The plots display changes in heart rate and pupil diameter prior to the takeover (pre-takeover period) and from the takeover (post-takeover period). The signals in blue were recorded during the monitoring condition, and in red during the MAI takeover condition. Shaded areas denote standard errors of the mean. Time points with significant differences between the monitoring condition and the takeover condition are displayed at the top of each figure as a red horizontal line. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 2
ANOVA outcomes comparing the MAI condition with the monitoring condition, in the pre- and post-takeover periods.

Measure	Period	Condition			Time			Condition x Time		
		F	(df)	n^2p	*F*	(df)	n^2p	*F*	(df)	n^2p
Heart rate	Pre-takeover	0.071	(1, 35)	0.002	0.61	(2.56, 89.84)	0.017	1.10	(2.23, 78.10)	0.031
	Post-takeover	0.047	(1, 35)	0.001	**11.6*****	(3.30, 115.7)	0.250	**8.2*****	(3.69, 129.2)	0.190
Pupil diameter	Pre-takeover	1.6	(1, 35)	0.046	1.0	(3.78, 132.4)	0.030	0.54	(3.37, 118.1)	0.015
	Post-takeover	**37.7*****	(1, 35)	0.519	**22.9*****	(2.61, 91.36)	0.396	**9.9*****	(3.88, 136.0)	0.221

Note. F (df) = statistics of ANOVA and degrees of freedom (in parentheses). * $p < .05$. ** $p < .01$. *** $p < .001$.

4. Discussion

In this study, the impacts of emergency takeovers on driver stress were explored using three takeovers involving three different levels of emergency. As expected, driver stress was greater for the takeovers performed in emergency situations (i.e. MDI and MAI conditions) than the one carried out in a non-emergency situation (i.e. ODI condition). Indeed, pupil dilation, higher levels of perceived stress and increased cardiac accelerations were observed for MDI and MAI conditions. Such results confirmed that time urgency is a key factor in the induction of driver stress during takeovers. As hypothesized, driver stress was reduced when the takeover could be anticipated (i.e. MAI condition) compared to when it could not (i.e. MDI condition). Indeed, for the MAI condition, lower perceived stress scores were reported and a cardiac deceleration before an acceleration was measured. These results support both the existence of a warning-induced preparatory response—characterized by a short-lasting decrease in heart rate and previously observed

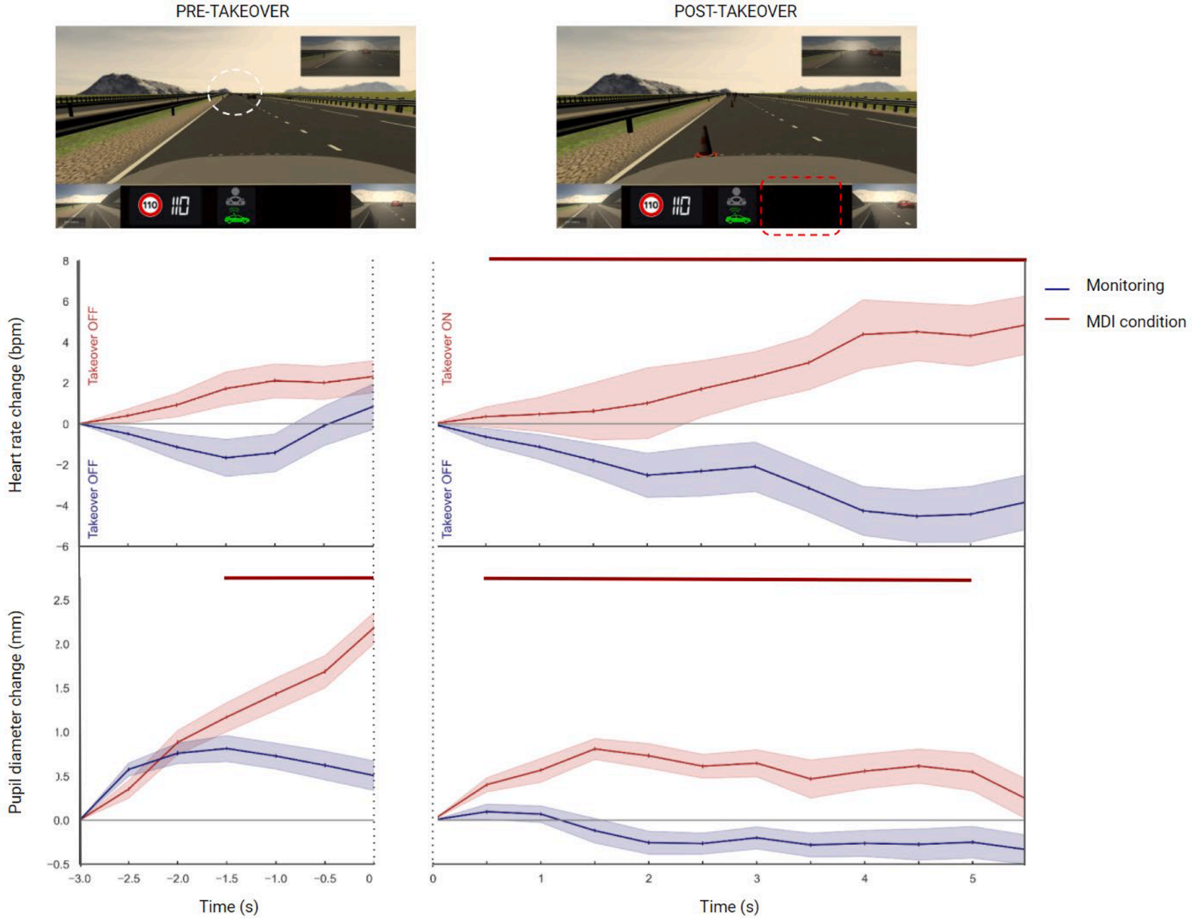

Fig. 6. The pictures depict the simulated environment 3 s before the takeover (left side), approaching a work zone (white dotted area), and at the moment when the takeover should have been requested (right side) with a warning (red dotted box). The plots display changes in heart rate and pupil diameter prior to the takeover (pre-takeover period) and from the takeover (post-takeover period). The signals in blue were recorded during the monitoring condition, and in red during the MDI takeover condition. Shaded areas denote standard errors of the mean. Time points with significant differences between the monitoring condition and the takeover condition are displayed at the top of each figure as a red horizontal line. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 3
ANOVA outcomes comparing the MDI condition with the monitoring condition, in the pre- and post-takeover periods.

Measure	Period	Condition			Time			Condition x Time		
		F	(df)	n^2p	*F*	(df)	n^2p	*F*	(df)	n^2p
Heart rate	Pre-takeover	2.0	(1, 35)	0.056	0.97	(1.97, 69.13)	0.027	2.6	(1.87, 65.48)	0.070
	Post-takeover	**23.2***	(1, 35)	0.399	0.90	(2.80, 98.21)	0.025	**10.5***	(2.40, 84.16)	0.231
Pupil diameter	Pre-takeover	**18.7***	(1, 35)	0.349	**39.0***	(2.68, 93.92)	0.527	**25.7***	(2.94, 103.0)	0.424
	Post-takeover	**15.6***	(1, 35)	0.309	**2.7***	(3.53, 123.5)	0.072	**5.0***	(4.24, 148.4)	0.127

Note. F (df) = statistics of ANOVA and degrees of freedom (in parentheses). * $p < .05$. ** $p < .01$. *** $p < .001$.

in research on defensive behaviors—and the hypothesis that the preparatory response is a means of reducing driver stress.

4.1. Impacts of takeovers on perceived stress

First, our findings indicated that regaining control in an emergency situation was perceived as more stressful than in a non-emergency situation. Indeed, higher stress levels were reported in the MDI and MAI conditions compared to the ODI condition. At this stage, one might suggest that only the different nature of driving situations (i.e. work zone, fog area, and toll area, respectively) between takeovers led to these results. However, we also found that MDI and MAI conditions were perceived as more stressful than monitoring conditions, i.e., conditions without takeovers but sharing the same driving situations as takeovers. Conversely, the ODI condition did not significantly differ in the perceived stress level from the monitoring condition. Taken together, these results suggest that driver stress during takeovers was not induced by a mere takeover, but by the urgency of the driving situation. This finding is consistent with the idea that limited time and insufficient preparation leads the driver to perceive more effort and uncertainty in performing all the tasks necessary to regain control, and thereby experience more stress (Ma et al., 2021; Wickens et al., 2015).

Second, the results in the emergency situations indicated that warning of a takeover reduced perceived stress levels compared to no warning of a takeover. Drivers indeed reported lower levels of stress in the MAI condition compared to the MDI condition. Again, the different nature of the driving situation (fog area and work zone, respectively) could not explain this stress reduction as no difference was retrieved between the two monitoring conditions (i.e. fog area vs work zone). According to these findings, the warning requesting drivers to regain control had a beneficial effect on driver stress by alleviating it during emergency situations. Recently, a research revealed that systems with two stages of warning (i.e. systems including an initial warning to inform driver to pay attention to potential hazards and a second one to require actions from the driver) reduced more efficiently driver stress, notably by reducing the levels of effort and uncertainty in performing takeover tasks successfully (Ma et al., 2021).

4.2. Impacts of takeovers on physiological stress

4.2.1. Takeover in a non-emergency situation (ODI)

The analyses indicated no change in heart rate between the ODI condition and the monitoring condition, before and after the takeover. These cardiac responses thus confirmed, at the physiological level, the similar low levels of stress reported by drivers in the non-emergency takeover and the monitoring conditions.

In addition, an increase in pupil diameter was observed in the ODI condition compared to the monitoring condition before the takeover and continued for a few seconds after the takeover. Such pupil dilation can be attributed to the detection and cognitive processing of the driving situation (here, the toll area). Conversely in the monitoring condition, drivers were informed well before the 3-seconds pre-takeover period of both the approach to a toll area and that the automation system would maintain control to cross it. Although, this additional information was provided to dissuade the participants from taking over in the monitoring condition, it may also have allowed drivers to detect and process the situation earlier than in the takeover condition, i.e., well before the 3-seconds-pre-takeover period. In other words, this additional information presented in the monitoring condition may explain the differences in pupil diameter between the monitoring and takeover conditions over the pre-takeover period. This interpretation is further supported by a previous research showing that increased pupil diameter is a good indicator to determine when an event or situation becomes relevant for awareness (Vintila et al., 2017).

4.2.2. Warned takeover in an emergency situation (MAI)

Before takeover, no change in heart rate was observed between the MAI condition and the monitoring condition. Such a finding demonstrated that drivers in both conditions experienced the driving situation (fog area) similarly prior to regaining control in one of the conditions. After being informed by the automation to resume control in the MAI condition, the drivers exhibited a strong and short-lasting decrease in heart rate, followed by a gradual and long-lasting increase in heart rate. This bi-phasic pattern, somewhat atypical at first glance, was widely observed in research addressing defensive behaviors during threatening situations including escape routes (Gladwin et al., 2016; Roelofs, 2017; Rösler and Gamer, 2019). Recently, this biphasic pattern was also observed in a driving simulator study in which the driver was alerted to an impending road hazard requiring an emergency avoidance maneuver (Keraudren et al., 2022). Building on the knowledge of defensive behaviors, we could interpret this bi-phasic pattern as follows. The strong cardiac deceleration observed after the warned takeover in the MAI condition would reflect attention and perception enhancement (Bradley, 2009). Furthermore, this cardiac deceleration would also reflect active preparation for a cognitive and/or motor action evidenced by subsequent cardiac acceleration (Gladwin et al., 2016; Rösler and Gamer, 2019). Using the lexicon of defensive behaviors, this would mean that the driver's short cardiac deceleration would be viewed as a freezing behavior while the driver's subsequent cardiac acceleration would be regarded as a flight behavior. Moreover, it should be noted that drivers' cardiac deceleration was observed during motor action as the drivers were taking control by pressing a pedal at the time of measurement. The fact that we did not observe a freezing behavior with a complete motor immobility, usually described in defensive behavior research, might be a generalization of traditional freezing behavior, as Ribeiro and Castelo-Branco (2019) previously suggested.

The current study also pointed out no change in pupil diameter between MAI condition and monitoring condition before the takeover. Indeed, pupil dilation was not expected for the monitoring condition as drivers were early informed about the driving situation (i.e. fog area detection) and the decision from the automation system to manage the driving situation. However, the fact that pupil dilation was not observed in the MAI condition before the takeover remains intriguing since the drivers had not received any information from the automation about the driving situation and its ability to manage the situation, unlike the monitoring condition. As previous research shown pupil dilation when drivers were stressed (Pedrotti et al., 2014) or when drivers detected road hazards (Vintila et al., 2017) and road hazard warnings (Bozkir et al., 2019), we can hypothesize that drivers did not judge the fog area as potentially stressful or hazardous, possibly explained by an overtrust in the automated driving at that moment. Additionally, a previous driving simulator study observed that pupil dilation decreased during driving in fog areas compared to non-fog areas due to the reduction in the contrast vision of the driving scene (Calsavara et al., 2021). We believe that the presence of fog in our scenarios might further explain our results, by disrupting the potential detection of pupil dilation. This hypothesis might also explain why pupil dilation was observed after but not before the warned takeover in the MAI condition. Indeed, the instruction to regain control, certainly responsible for the dilation after takeover in the MAI condition, was displayed on the dashboard and thus was not affected by the fog. Finally, the great and long-lasting pupil dilation observed after takeover in the MAI condition was consistent with the idea that the situation was sufficiently relevant for awareness (Vintila et al., 2017.

4.2.3. Non-warned takeover in an emergency situation (MDI)

When drivers had to take control from the automation system in an emergency situation (MDI condition), the current study found an increase in heart rate compared to when drivers remained in a monitoring activity. The immediate increase in heart rate suggests that a behavior similar to the flight behavior was adopted. The lack of cardiac deceleration preceding cardiac acceleration in the takeover condition was demonstrated in the pre- and post-takeover periods. According to the defensive behavior theory, such a result would indicate that drivers were not able to prepare and anticipate an adequate response to the driving situation. Indeed, numerous research has demonstrated a relationship between active preparation (or anticipation) under threat imminence and cardiac deceleration (Gladwin et al., 2016; Hashemi et al., 2019; Löw et al., 2015; Rösler and Gamer, 2019). In addition, the lack of cardiac deceleration in drivers could be explained by the fact that they did not receive a warning from the automation system requiring them to take control or informing them of the approaching hazard. Therefore, the drivers initiated the control transition late, i.e. at the moment they became aware of the automation system's inability to properly handle the situation. This failure from the system was revealed to the drivers by the fact that the vehicle was too close to the road cones or upon collision with the cones. Overall, this cardiac response suggests that the lack/reduction of driver anticipation in extreme emergency driving situations could be evidenced physiologically by the absence of cardiac deceleration preceding acceleration.

Interestingly, while heart rate increased after resuming control in the MDI condition, it decreased in the monitoring condition. This long-term cardiac deceleration corresponded to the time when drivers were monitoring the situation while the automation system correctly identified a work zone and directed the vehicle onto the other lane. In this work zone were highway maintenance vehicles and workers. The long-lasting cardiac deceleration found in this potentially hazardous situation converges with previous research addressing defensive behaviors (Kastner-Dorn et al., 2018). Indeed, Kastner-Dorn and colleagues noticed a long-lasting cardiac deceleration in an anxiety-inducing context that they associated with hypervigilance. The authors also indicated that such behavior would facilitate sustained processing of sensory information. In addition, vigilance has been equated with risk assessment and with ambiguity of threat stimulus (Blanchard et al., 2011). According to the theoretical basis of defensive behaviors, the long-duration cardiac deceleration observed in drivers would reflect heightened attention and scanning for potential road hazard and/or failure of the automation system.

Furthermore, an increase in pupil diameter was observed in the MDI condition compared to the monitoring condition before the takeover and continued after the takeover throughout almost the entire post-takeover period. The fact that pupil dilation was observed before takeover without a significant increase in heart rate suggests that drivers detected a relevant situation (probably triggered by the presence of road cones), but that it did not trigger a stress response at that time. In contrast, after regaining control, the pupil dilation associated with an increase in heart rate indicates that the situation has finally been experienced as stressful.

4.2.4. Synthesis: Warned versus non-warned takeover in emergency situations

As mentioned above, drivers reported less stress in emergency situations for the warned takeover (MAI) than for the non-warned takeover (MDI). The most likely hypothesis to explain this difference is that the warning from the automation system gives the driver additional time to prepare takeover-related tasks. This hypothesis is also supported by the exploration of cardiac patterns interpreted in light of the literature on defensive behavior. Indeed, a preparatory response was deduced from cardiac deceleration in the warned emergency takeover while no such response was spotted in the non-warned emergency takeover. For the latter takeover, only a cardiac acceleration was noticed.

4.2.5. Hypothesis: Hypervigilance versus selective attention

In emergency situations, cardiac decelerations were spotted after the warned takeover (MAI condition) and during a monitoring condition including a potentially hazardous driving situation (i.e. work zone). Specifically, a short-lasting cardiac decrease was observed for the MAI condition while a long-lasting cardiac decrease was noticed for the monitoring condition involving a long work zone. According to the defense cascade model (Fanselow, 1994; Lang et al., 1997), the cardiac decelerations observed in our study would be explained by the proximity (or distance) with the hazard. Schematically, the long-lasting decrease in heart rate would correspond to the post-encounter phase, when a hazard is ambiguous or far. Conversely, the short-lasting decrease in heart rate, followed by a subsequent increase in heart rate, would be broadly equivalent to the circa-strike phase, when hazard was proximal or imminent. Attentional mechanisms accompanying these two stages were disentangled in a previous study (Kastner-Dorn et al., 2018). Kastner-Dorn and colleagues found hypervigilance in the post-encounter phase while selective attention was highlighted when the threat was imminent (circa-strike stage). Therefore, the defensive cascade model seems to be suitable to explain drivers' behaviors during emergency takeovers. In particular, using this model would provide substantial theoretical insight into the cognitive processes involved in the preparatory response (anticipation) as well as attentional mechanisms (hypervigilance and selective attention). In addition, further research would confirm or refute the hypothesis that a cardiac signature would provide an understanding of the levels of attention and vigilance involved in a monitoring activity, especially during highly automated driving.

4.3. Limitation

A first limitation of the study is that all conclusions regarding the takeover and monitoring conditions were based on distinct driving situations (i.e., toll, fog, works). In order to generalize the findings, it would be useful to explore defensive behaviors while driving in other situations than those used in the present study. It should be noted that since each of the three types of takeover explored here (i.e. ODI, MAI and MDI) corresponded to different degrees of emergency, it was not feasible to implement all three in the same driving situation.

A second limitation of the study concerns the metrics used. Indeed, both heart rate and pupil diameter can be sensitive to various internal and external factors that can modulate their responses. In order to minimize some of these factors, the experiment was conducted in a completely dark room to avoid changes in light, which are known to impact pupil diameter (Pfleging et al., 2016). In addition, to avoid any heart rate disturbances, participants were asked not to consume energy drinks or sugar in the 2 h before the experiment. Age and gender were two factors that were controlled as much as possible. Although the participants are represented over a relatively wide age range, it should be noted that men are outnumbered by women.

A third limitation refers to the simulated driving environment. Although it is supposed to be as close as possible to real driving, it does not recreate the same vital concerns and driving sensations. Since these differences in environment can have a significant impact on drivers' responses, it would be interesting in future studies to compare drivers' responses on real roads and in driving simulators in order to generalize the results found in this study beyond the simulated environment.

5. Conclusion

The current study has provided a better understanding of the impacts of emergency takeovers on driver stress. In addition, our findings suggest that the use of the classification of transitions developed by Lu and colleagues (2016) combined with a theoretical approach of defensive behaviors offer, together, a promising theoretical framework to gain insight into driver's cognitive processes involved in automated driving.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.trf.2023.01.016.

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